Srijan Tiwari

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SUMMARY

Final-Year B.Tech Undergraduate at **IIT Guwahati** with Machine Learning and Data Science experience leading a **6-member** group at **Genuity IO**. Strong foundation in Deep Reinforcement Learning, Computer Vision, LLM Evaluation, and Information Retrieval across **100,000+** records.

EDUCATION

Indian Institute of Technology (IIT), Guwahati — *Bachelor of Technology (B.Tech)* **Aug 2023 - May 2027** | *Guwahati, India*
*Board Examinations: CBSE Class XII: **84.4%** | CBSE Class X: **94.6%**.*

EXPERIENCE

Genuity IO — *Lead Software Engineer, Synthetic Data and Generative AI* **May 2026 - Jul 2026** | *Bengaluru, India*

- **Architected** Genuity **v2.0**, a synthetic dataset generation library supporting **5 modalities** (tabular, time-series, financial, relational, and text) and **20+ generative models** (CTGAN, TVAE, and TabDDPM), driving production integration into the Vulpix platform.
- **Led** a cross-functional group of **6 engineers** across architecture design, code reviews, and sprint delivery; established a benchmarking framework to quantify statistical fidelity, downstream ML utility, and confidentiality across **20+ models** and **11 public benchmarks**.
- **Engineered** production backend workflows for distribution profiling, constraint validation, and differential privacy using **Python, FastAPI, Docker, and PostgreSQL**, scaling the prototype to enterprise client deployment for **L&T Technology Services**.

PROJECTS

Promo Finder: AI Video Script-to-Promo Clipping Pipeline | *Python, MiniLM, Gemma-26B, Streamlit* **Oct 2026 - Nov 2026** | GitHub

- **Developed** a hybrid rule-based and Generative AI workflow that parses timestamped screenplays into structured **Pydantic** scene beats, extracting standalone **20-to-60-second** promotional clips with engaging **3-second** opening hooks and clean dialogue cuts.
- **Modeled** a **0.25s** dramatic intensity score across each script by combining pacing cues (camera cuts, speaker transitions, and sound effects) with **Sentence-Transformers (MiniLM)** emotional tone embeddings to locate peak scenes.
- **Constructed** a cold-viewer **LLM-as-a-Judge (Gemma-26B)** to score candidate clips on hook strength, cliffhanger tension, clarity, and spoiler risk across **4 evaluation metrics**, paired with a story-arc montage builder, **Streamlit** UI, and **Pytest** suite.

Algorithmic Trading via Deep Reinforcement Learning | *Python, TensorFlow 2, DDDQN, BiLSTM, Gymnasium* **Jan 2026 - Feb 2026** | GitHub

- **Trained** a **Dueling Double Deep Q-Network (DDDQN)** with a **Bidirectional LSTM** encoder, analyzing **1,820+ historical Bitcoin daily bars (2020 - 2025)** across **10+ technical indicators** and a **90-day** lookback window to decouple value and advantage streams.
- **Designed** a position-aware **Gymnasium** trading environment with **3 discrete actions** (Buy, Sell, and Hold), simulating realistic market friction via **0.1% (10 bps)** transaction fees, dynamic position sizing, stop-loss triggers, and drawdown controls.
- **Accelerated** training convergence to **500 episodes** by implementing **Prioritized Experience Replay (PER)**, combining temporal-difference error sampling, importance-sampling bias correction, **64-sample** mini-batches, and **20-step** target network syncs.

Large-Scale Candidate Ranking and Trap Detection System | *Python, TF-IDF, Bounded Heaps, Gradio* **May 2026 - Jun 2026** | GitHub

- **Streamlined** a single-pass information-retrieval workflow that evaluates **100,000 candidate records (487 MB)** in under **65 seconds** within **120 MB of peak RAM**, utilizing bounded min-heap **Top-100** selection (**O(N log K)**) and **TF-IDF** cosine similarity.
- **Formulated** a **20-feature** scoring model and anomaly filter across **70,584** eligible applicants to flag fabricated histories, timeline inconsistencies, and keyword stuffing, significantly reducing false positives in the ranked output.
- **Created** a deterministic natural-language generator that produces **480** structured, evidence-backed recruiter justifications via hash-based template selection, deployed through an interactive **Gradio** interface on **Hugging Face Spaces**.

Semantic Image Segmentation with Nested U-Net++ | *Python, Deep Learning, U-Net++, NumPy, Kaggle GPU* **May 2025 - Jun 2025** | GitHub

- **Built** an optimized **U-Net++** encoder-decoder architecture (**32 to 512 filters** with nested skip connections and dropout regularization) for pixel-wise semantic segmentation on RGB image-mask pairs, mitigating class imbalance via combined **Dice and Cross-Entropy Loss**.
- **Implemented** a vectorized **NumPy** RGB-to-class-ID label mapping module and morphological refinement (dilation and erosion) to sharpen boundary masks, reaching **80% validation accuracy** and **230 ms/image** inference latency within a **395 MB** checkpoint.

MolConvert: 3D Molecular Coordinate Reconstruction CLI | *Python, NumPy, RDKit, Biopython, Pytest* **Jun 2026 - Jul 2026** | GitHub

- **Programmed** a structural bioinformatics CLI that translates across **9 molecular file formats** (including PDB, SDF, Gaussian, and Z-Matrix) via a unified JSON internal-coordinate representation (bond length, bond angle, and torsion angle).
- **Executed 3D** coordinate reconstruction using **NumPy** rotational matrices with low round-trip RMSD deviation, Kabsch structural alignment, **RDKit** chemical validity checks, and **376 Pytest unit tests**.

SKILLS

Programming Languages: Python, C++, C, SQL, JavaScript, Solidity, Bash, LaTeX.
Machine Learning and Deep Learning: PyTorch, TensorFlow 2.x, Keras, Scikit-Learn, Deep Reinforcement Learning (DDDQN, DQN, PER, Gymnasium), Computer Vision (U-Net++, CNNs), Sequence Modeling (BiLSTM, Transformers).
Generative AI, NLP and Retrieval: LLM Orchestration (Gemini, OpenAI, Groq, OpenRouter), LLM-as-a-Judge Evaluation, Sentence-Transformers (MiniLM), TF-IDF Vectorization, Cosine Similarity, Prompt Engineering, Structured Outputs (Pydantic).
Data Science, Bio-ML and MLOps: NumPy, Pandas, Matplotlib, SciPy, RDKit, Biopython, Differential Privacy, Synthetic Data (CTGAN, TVAE, TabDDPM), FastAPI, Streamlit, Gradio, Docker, Git, PostgreSQL, Pytest, Hugging Face Spaces, Kaggle GPU.
Mathematics and CS Coursework: Linear Algebra, Multivariable Calculus, Probability and Statistics, Differential Equations (ODE/PDE), Hardware-Aware Deep Learning, Data Structures and Algorithms, Bioinformatics and Computational Biology, Quantum Computing.

ACHIEVEMENTS AND LEADERSHIP

- **Ranked** in the **Top 10 Percentile** among **1,000+ participants** nationwide in the premier **Summer Analytics** Machine Learning and Neural Networks competition (**Jul 2024**, Certificate); completed **Product Matters 4.0** (Certificate).
- **Selected** among the **1st delegation from IIT Guwahati** to represent the institute at the **ETHGlobal New Delhi Hackathon (Oct 2025)**, delivering Web3 protocol infrastructure during a **36-hour** international sprint.
- **Spearheaded** quantitative finance and blockchain modules as **Executive Member** of the Finance and Economics Club (**Mar 2025 - Apr 2026**), mentoring **200+ students** across algorithmic trading and cryptography sessions.
- **Elected** to the Hostel Management Committee (**Apr 2025 - May 2026**), overseeing operations for **400+ residents**; wrote and directed a short film, winning **4th Place** campus-wide (Video).